

National University of Computer and Emerging Sciences

Fast School of Computing

Spring 2026

CS-4031 Compiler Construction

Quiz # 4 [20 marks]

Section: _____

Time = 20 minutes

07/04/2026

Name: _____

Roll No: _____

Question 1: Fill in the blanks.

1. If $S \rightarrow CC$, $C \rightarrow cC$, $C \rightarrow d$ then $I_0 : \text{closure}(\{(S'' \rightarrow \bullet S, \$)\}) = \underline{\{[S \rightarrow \bullet CC, \$], [C \rightarrow \bullet cC, c|d], [C \rightarrow \bullet d, c|d]\}}$
2. If $[A \rightarrow B, a]$ is a production in LR(1) items, then reduce operation is entered under a.
3. LR(1) items are used for LALR(1) parsers.
4. The type of item used in $[A \rightarrow \bullet Aa, a|b|c]$ LR(1).
5. Relation between grammars LL(1) is a subset of LL(k).
6. Relation between grammars LL(1) is a subset of LR(k).
7. If there are no RR conflicts in a LR(1) parser then it may occur in LALR(1).
8. If there are no SR conflicts in a LR(1) parser then it will not occur in LALR(1).
9. The other name for bottom-up parser is shift-reduce parser.
10. $I_0 : [S \rightarrow \bullet aA, \$] [A \rightarrow \bullet \epsilon, a]$ the state I_0 has S/R conflict.

Question 2: Consider the following grammar.

1. $G \rightarrow P$
2. $G \rightarrow PG$
3. $P \rightarrow id : R$
4. $R \rightarrow \epsilon$
5. $R \rightarrow id R$

Construct the DFA of LR(1) items for this grammar.